Jordan River Project Water Use Plan

Lower Jordan River Salmon Spawning Assessment and Enumeration

Reference: JORMON#3

Jordan River Spawning Assessment/Enumeration 2005/2006

Study Period: Sep 05-May 06

Cascadia Biological Services

1442 White Pine Terrace

Victoria, BC

V9B 6J3

Executive Summary

Cascadia Biological Services was retained by BC Hydro, Campbell River to perform a salmon enumeration and incubation study as part of the Jordan River Water Use Plan Monitoring Program. The study area encompassed both Reach 1 and Reach 2 of the lower Jordan River between the tailrace of the powerhouse and an impassable barrier located approximately 590m upstream.

The overall goal of this project was to determine/enumerate salmonid spawning focusing on 2 different salmonid runs including Coho salmon from September 21st to December 21st 2005 and for Steelhead from April 15th to May 31st 2006. A secondary goal was to perform 5 incubation tests in areas of high spawning suitability within the defined study area to determine incubation success of each developmental stage.

The results of the enumeration study show that adult salmon and steelhead returns to Lower Jordan River were extremely low with only 4 individuals recorded over eight stream counts. The Coho spawning surveys witnessed 4 individuals over 5 counts in the fall, while no steelhead were observed over three counts in the spring.

The results of the incubation study show that incubation successes were high (94%) at 4 out of the 5 incubation resulting in an overall mean survival to stage sampled of 76% for all five sites. Spawning gravel within the study area was generally considered to be of low-moderate quality with approximately 144m^2 available within the entire study area. Of the 144m^2, it was determined that 118m^2 (82%) was available to spawners during low to moderate flows. The remaining 26 m^2 would be available during high to extremely high flow events.

Results to date show potential limits to spawning success in the Lower Jordan River due to lack of spawning habitat (flow, gravel quality and quantity) and water quality issues from historic mine abandonment. These results are preliminary and require further annual assessments to confirm.

Table of Contents

Executive Summary 2

Table of Contents 3

List of Figures 4

List of Tables 4

1.0 Introduction 5

1.1 Project Area 5

1.2 Objectives of the Work 9

2 Methodology 10

2.1 Survey Preparation 10

2.2 Fall and Spring Surveys 10

2.3 Winter/Spring Incubation Tests 10

3 Results 20

3.1 Study Area Delineation and Description 20

3.2 Spawning Assessment Results 2005/2006 22

3.3 Incubation Assessment 27

3.3.1 Incubation Site Summary 28

3.4 Spawning Gravel Assessment 33

4.0 Summary and Discussion 36

5.0 Recommendations 37

6.0 References 38

List of Figures

Figure 1 Jordan River Study Area 7

Figure 2. Mean Monthly Discharge for Jordan River (1984-1999)..... 8

Figure 3. Incubation Site Map 13

Figure 4. Spawning Assessment Map 26

Figure 5. Spawning Gravel Assessment Map..... 34

Figure 6. Spawning Gravel Assessment Map..... 35

List of Tables

Table 1. Incubation Site Summary at Time of Placement 12

Table 2. Reach 1 Biophysical Delineation..... 21

Table 3. Reach 2 Biophysical Delineation..... 22

Table 4. Spawning Assessment Summary 24

Table 5. Incubation Assessment Summary 2006..... 31

Table 6. Incubation Assessment Summary (Pink Salmon) – Wright and Guimond 2003 . 32

1.0 Introduction

BC Hydro and Power Authority initiated a Water Use Plan (WUP) for the Jordan River watershed in April 2000 and concluded it in November 2001. From this WUP, a minimum flow was ordered for fisheries benefits, and a 6-year monitoring plan was established to evaluate the effectiveness of the ordered flow regime. The Lower Jordan River Spawning Assessment and Enumeration study program is one component of the monitoring plan.

The overall objective of the study program is to assess the effectiveness of the minimum flow order for Lower Jordan River using spawning success as a key indicator. Spawning success is defined as the number of recruits per spawner within the anadromous reaches (approximately 590m in length) (refer to Figure 1). Target species were coho salmon (surveyed September 21st to December 21st 2005) and steelhead trout (surveyed April 15th to May 31st 2006). Study objectives were pursued through snorkel surveys and incubation assessments within the anadromous section of river (reaches 1 and 2).

1.1 Project Area

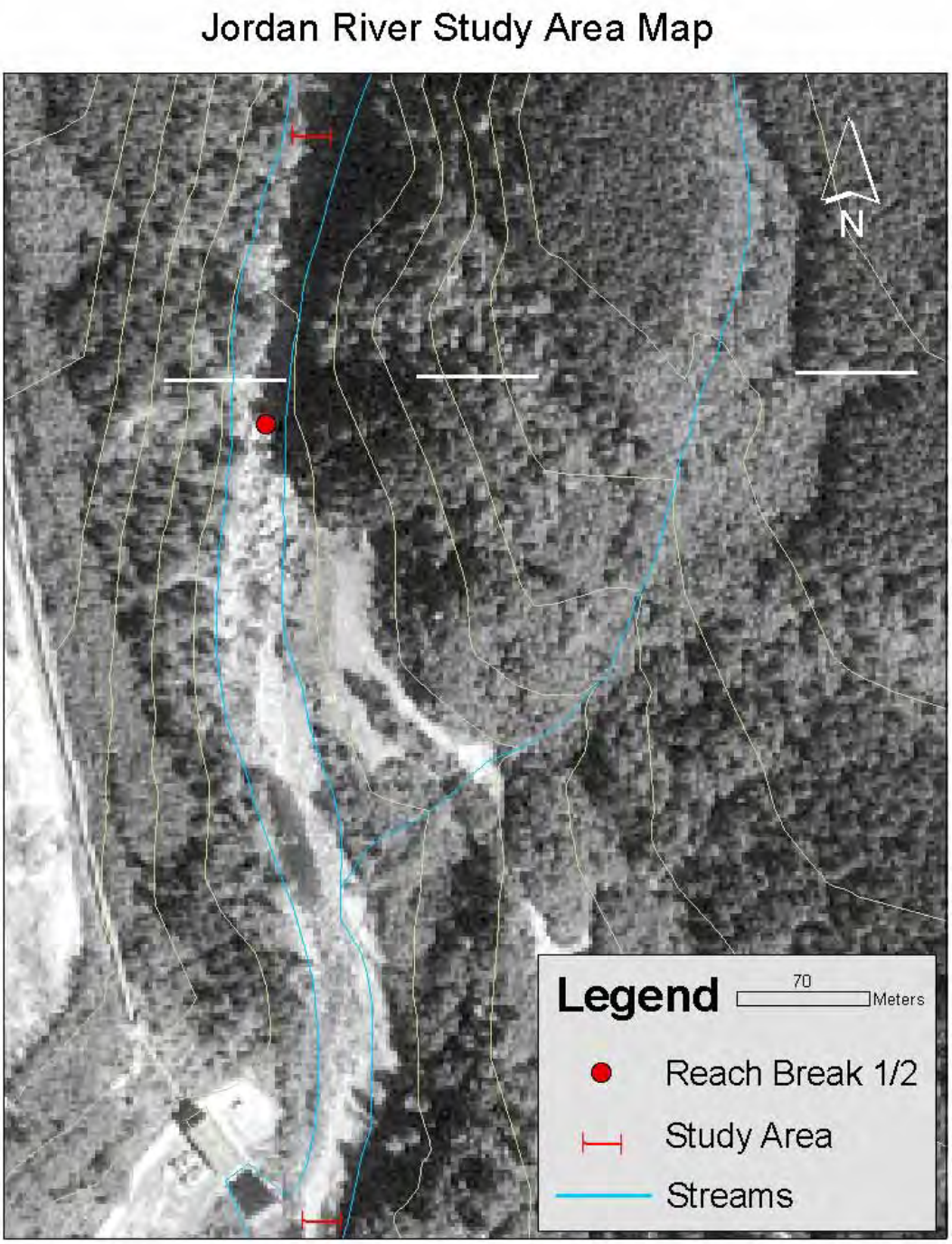
The study area (reaches 1 and 2 from Figure 1) encompasses the mainstem of Jordan River (WSC- 930-037300) between the powerhouse tailrace and an impassable barrier to upstream fish migration located approximately 590m upstream.

The Jordan River drainage is located on the west coast of southern Vancouver Island, approximately 60 km west of Victoria. The river empties into the Juan de Fuca Strait. The existing river habitat is approximately 25 km long (excluding two reservoirs- Diversion and Elliot), of which approximately 7.4 km is between Elliott Dam and the powerhouse tailrace. The Lower Jordan River downstream of the powerhouse tailrace is tidally influenced and is not part of the study area.

The drainage area of Jordan River is approximately 184 km^2, of which 80% (147 km^2) is captured at the power intake at Elliott Dam. Under normal operating conditions, inflows and storage at Elliott Reservoir are diverted to the Jordan River powerhouse via a tunnel and penstock. Flows in the lower Jordan River channel below Elliott Dam are derived from local input from the residual 37 km^2 of the watershed.

Where inflows to the Elliott Reservoir exceed its storage, BC Hydro will release flows from Elliott Dam to avoid uncontrolled spill events.

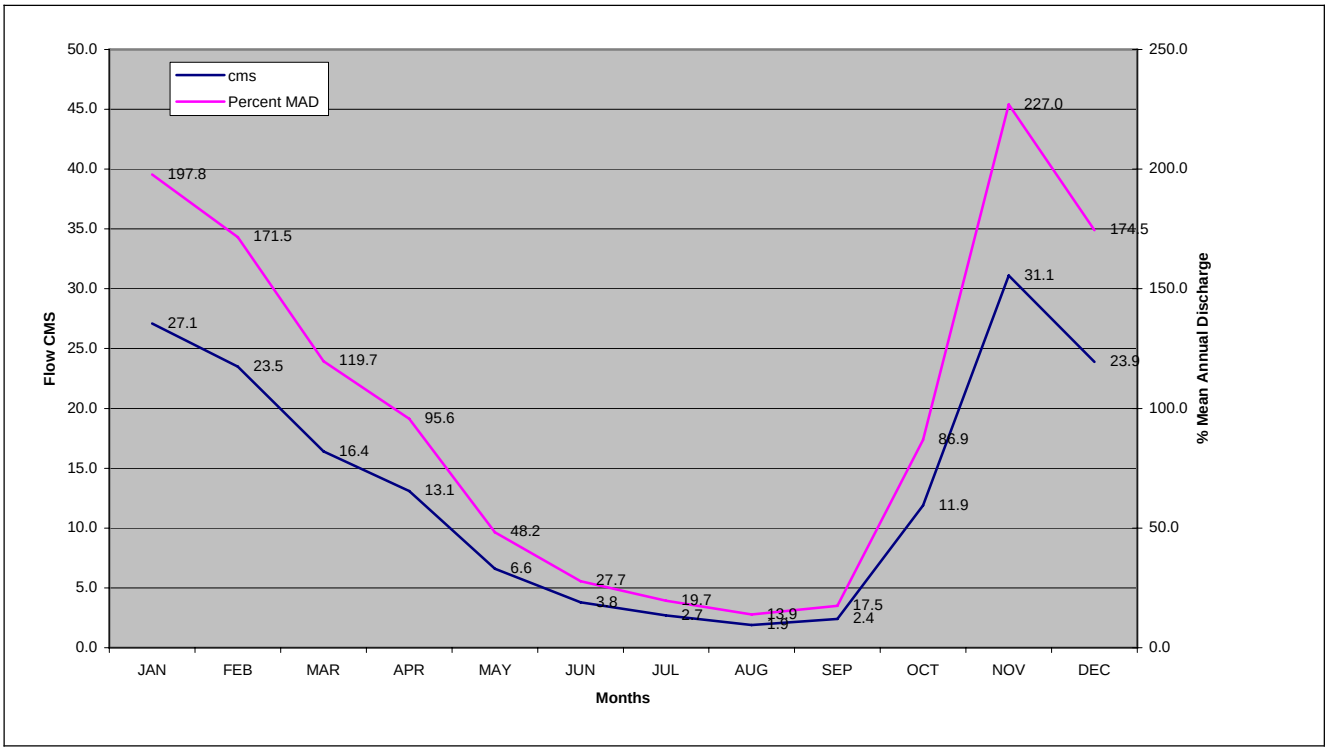
Figure 1 Jordan River Study Area



The mean annual precipitation within the watershed is 2,830 mm. The heavy precipitation events are usually limited to the winter period (October- March). Active logging within the watershed has reduced its capacity to attenuate precipitation events. Consequently peak flows during the winter have been more severe, while in the summer months periods with reduced or no inflow are frequent. Other factors affecting fish spawning activities within the study area include construction of the new tailrace and generating facility, past mining operations and an active log sort at the Jordan River confluence.

Based on daily Lower Jordan River inflows calculated for the 1984-1999 period, the mean annual discharge is 13.7cms (BC Hydro 2001) measured at a gauging station located approximately 125m upstream of the tailrace. The mean monthly discharge calculated using this data set is shown in Figure 2.

Figure 2. Mean Monthly Discharge for Jordan River (1984-1999).



A FISS (Fish Information Summary System) report generated June 2, 2006 indicated the presence of anadromous cutthroat trout, chum salmon, coho salmon, cutthroat trout, rainbow trout, and steelhead.

1.2 Objectives of the Work

The following objectives were initially identified (BC Hydro 2005):

The overall objective of this monitoring program was to assess the performance of the key WUP decision to increase flows in the lower Jordan River from leakage/local inflows to >0.25cms using spawning success (number of returns to outmigrants) as the performance measure. As a requirement for establishing baseline performance measures, specific objectives related to our monitoring program included the following;

- Evaluate incubation success (Coho salmon) at five specific sites within the delineated study area;
- Determine the number of spawning adults within the study area during specific timing/spawning windows for both Coho salmon (September 21st – December 21st) and for the Steelhead (April 15th – May31st).
- Determine spawning gravel quantity and quality within the study area.

2 Methodology

2.1 Survey Preparation

Survey preparation for the project included the following;

- A detailed review of past projects on the lower Jordan River
- Assess safety concerns/site conditions with BC Hydro staff prior to the initiation of snorkel surveys.
- Produce a detailed safety plan and have it reviewed by BC Hydro staff.
- Develop a detailed map of spawning/incubation habitat.

2.2 Fall and Spring Surveys

A total of eight fall and spring snorkeling surveys were completed during both the Coho spawning period (September 21st 2005 to December 21st 2005 – 5 surveys) and the Steelhead spawning period (April 15th 2006 to May 31st 2006 – 3 surveys). Due to safety concerns, a crew of three was utilized, whereby two snorkelers were in the water at all times and the third crewmember traveled onshore parallel to the two swimmers. The onshore crewmember maintained constant contact via two-way radios to the powerhouse and was equipped with various flotation devices as well as towropes. Where possible, the river was snorkeled/assessed twice from top (start point) to bottom. An underwater camera was used to photograph spawners. Staff gauge and water temperature measurements were also taken during each survey as well as select photographs documenting river conditions at the time of survey.

2.3 Winter/Spring Incubation Tests

Eyed coho salmon eggs (*Oncorhynchus kisutch*) were obtained from the Jack Brooks Hatchery in Sooke, BC, at approximately 256 ATU (accumulated thermal units) and transported to Jordan River on January 20th, 2006. Prior to transportation, staff at the hatchery sanitized all equipment including a 15-gallon tote with a diluted iodine solution. Eggs were then placed in the tote with water from the hatchery and covered with a sterilized cloth to reduce impacts associated with travel. The total time elapsed between

the hatchery and insertion into the cassettes ranged from 1-2 hours. Once at the incubation site locations, eggs were manually inserted into the cassettes (200/cassette - Jordan-Scotty incubation cassettes). The cassettes were then zip tied together forming three cassettes (600 eggs) per site. Cassettes were then planted in the gravel (holes dug one week prior to insertion), at five different sites in the river. Once planted, the cassettes were covered with gravel and tied to a piece of rebar vertically implanted into the streambed. Three of the five sites (Sites 3,4 and 5) replicated incubation sites utilized in a incubation study completed in 2003 (Wright and Guimond, 2003). The locations and biophysical summaries of the incubation sites are summarized in Figure 3 as well as in Table 1 below.

Due to technical difficulties with the incubation cassettes, individual plates at each incubation site could not be separated from each other once buried and therefore resulted in the entire site being pulled for each stage sampled rather than one incubation cassette being pulled from each site at each stage of incubation.

Egg survival was assessed at hatch, emergence as well as at post emergence at select sites discussed further in the results section. After each assessment and where possible, eggs/alevins were buried in the gravel once assessed for stage and health.

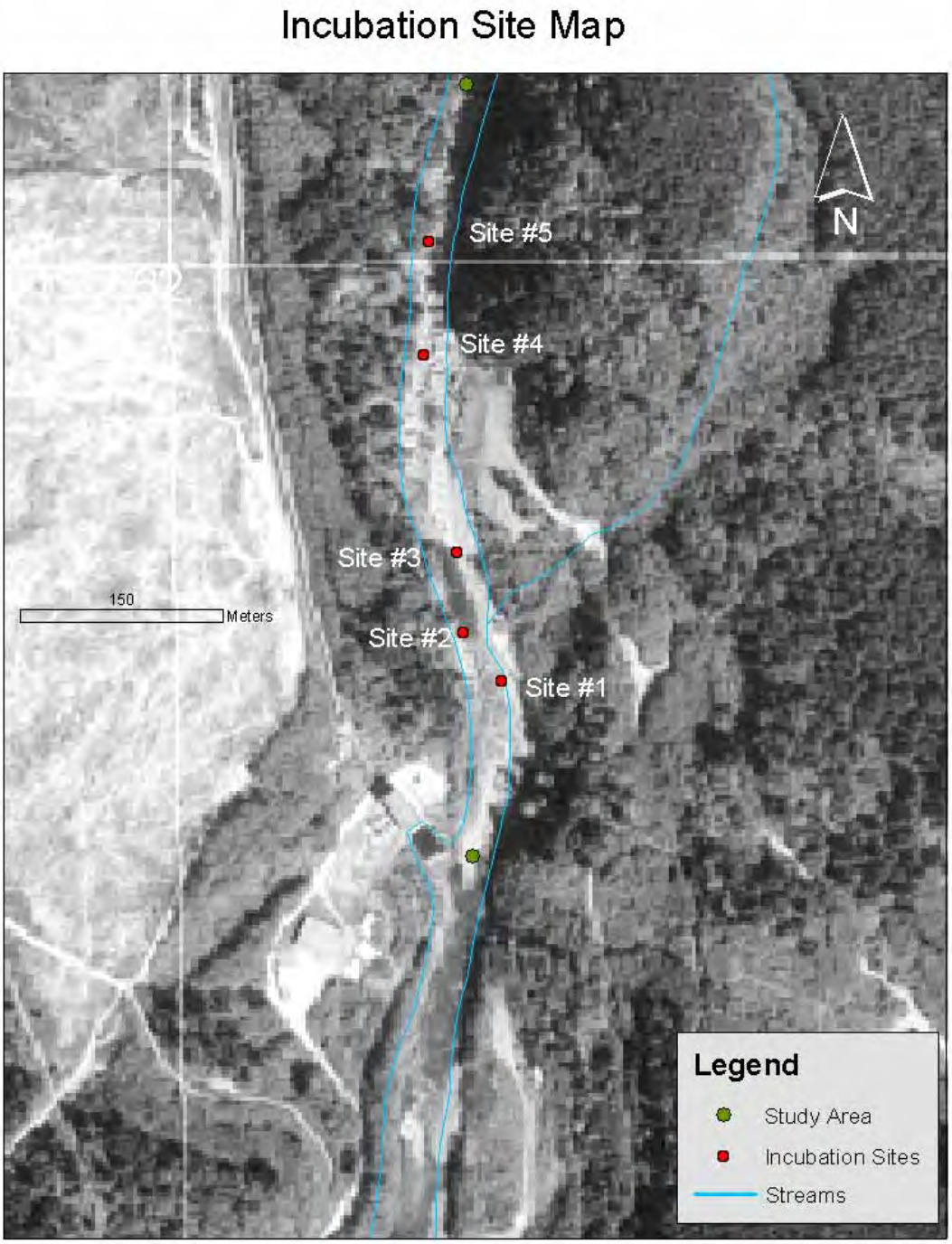
Water temperature was monitored at each visit to the assessment area using a handheld mercury thermometer.

Stage was monitored by BC Hydro's gauging station located approximately 125m upstream of the tailrace and correlated to flow using the stage-discharge relationship for the station generated by Hydroid Geoscience Ltd (in prep).

Table 1. Incubation Site Summary at Time of Placement

Incubation Site #	Water Depth Over Cassettes	Water Velocity at 60% Depth	Temp.	Substrate	Total Dissolved Oxygen (Time of Placement) (mg/l)	Silt Content In Cassettes When Sampled
1	0.32m	0.11m/s	6°C	Gravel (20-65mm)	10.65	No Silt
2	0.41m	0.08m/s	6°C	Gravel (15-45mm)	9.36	No Silt
3	0.38m	0.071m/s	6°C	Gravel (10-25mm) and sand	8.25	Some Silt in Cassettes
4	0.21m	0.11 m/s	6°C	Gravel (20-40mm)	10.31	No Silt
5	0.39m	0.91 m/s	6°C	Gravel (10-30mm)	10.48	No Silt

Figure 3. Incubation Site Map



The following photographs present typical river conditions at time of egg placement into gravel. Where possible, views of upstream/downstream conditions as well as substrate composition at each incubation site were photographed. All photos were taken on January 20th, 2006.

Site #1 – Glide section of last riffle before marine influence (Plates #1 and 2)

Plate 1. Typical view of Site #1 facing upstream,

Plate #2. View of substrate (Site 1)

Site #2 – Tailout of large pool opposite mine tailings (Plates 3 and 4)

Plate #3. Side view of Site #2.

Plate #4. View of substrate (Site #2)

Site #3 – At outflow (3m u/s) of large tributary coming from tailings (Plates 5 and 6)

Plate #5. Upstream view of Site #3.

Plate #6. View of substrate (Site #3)

Site #4 – 40m d/s of cascade reach (Plates 7 and 8)

Plate #7. Downstream view of Site #4.

Plate #8. View of substrate (Site #4)

Site #5 – 75m u/s of Site #4 in cascade habitat (Plates 9 and 10)

Plate #9. Side view of Site #5.

Plate #10. View of substrate (Site #5)

3 Results

3.1 Study Area Delineation and Description

Reach 1

General channel characteristics of Reach 1 included an average channel width of 43.20m and average gradient of 3%. The stream morphology most common within Reach 1 consists of riffle-pools with large boulders being the dominant substrate and a mix of small and large cobble sub-dominant. Remnants of an abandoned mine lie on the left bank of the river. Seepage from the mine contributes to elevated levels of toxic heavy metals (Griffith 1996). Refer to Figure 1 for reach delineation. Sub-delineation of Reach 1 into smaller biophysical units is presented below in Table 2

Table 2. Reach 1 Biophysical Delineation

Section	Location	Dominant Substrate	Spawning Area (m^2)	Primary Cover
1	0-171m	boulders/cobble	44	boulders
2	171-249	cobble	30	deep pool
3	249-419	boulders	45	boulders

Fish habitat characteristics of Reach 1 include good deep pool and boulder cover. LWD cover within the reach is considered poor. Pockets of spawning gravel exist intermittently throughout the reach and consist of large gravel/small cobble sized substrate. A typical view of the habitat within the reach is shown in photographs below:

Reach 2

General channel characteristics of Reach 2 included an average channel width of 31.80m and average gradient of 18%. The reach extends up from Reach 1 into higher gradient cascades to an impassable barrier located approximately 590m u/s of the tailrace. Refer to Figure 1 for reach delineation. Sub-delineation of Reach 2 into smaller biophysical units is presented below in Table 3

Table 3. Reach 2 Biophysical Delineation

Section	Location	Dominant Substrate	Spawning Area (m^2)	Primary Cover
1	419-435m	boulders/cobble	20	deep pool
2	439-590	boulders/cobble	5	boulders

Fish habitat characteristics of Reach 2 include good deep pool and boulder cover. LWD cover within the reach is considered poor/non-existent. Small pockets of spawning substrate exist occasionally throughout the reach and are generally located at pool tailouts and consist of small cobble sized substrate. A typical view of the habitat within the reach is shown in photographs below:

3.2 Spawning Assessment Results 2005/2006

Escapement results from the 5 coho surveys and 3 steelhead surveys are summarized in Table 4 below. Four coho spawners and no steelhead were observed during the surveys. In all cases, surveys were attempted during high water events when conditions would be more favorable. All four coho spawners were observed over a period of two weeks between October 20th and November 3rd, 2006 and were all located upstream of the large pool adjacent to the mine tailings and below the steep cascade section. One of the four individuals however, was observed in a seal's mouth immediately below the powerhouse tailrace. Of the three individuals observed in the river channel, one female (adipose unclipped, presumably wild) was identified and had obvious scarring on her caudal fin

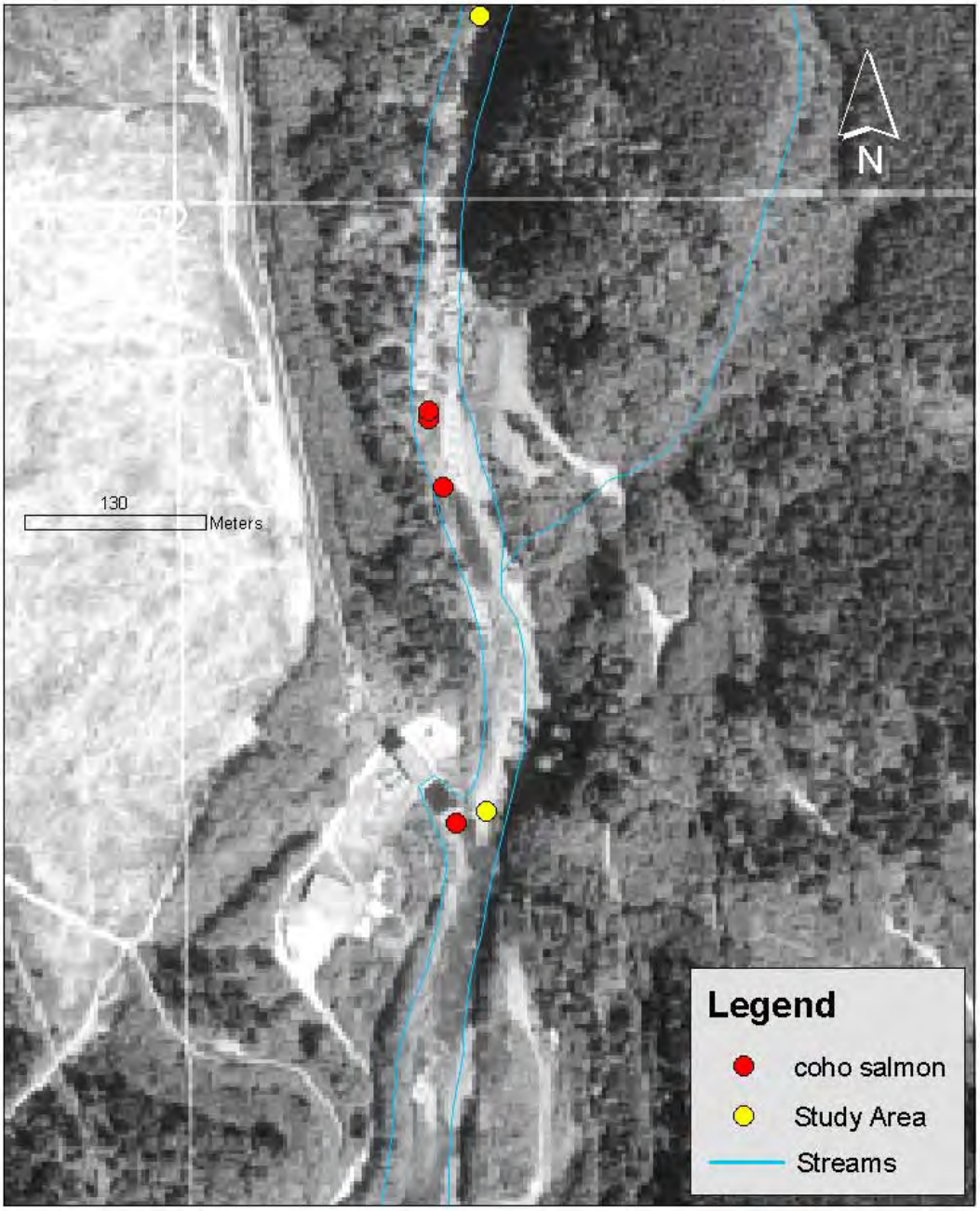
suggesting redding had occurred. Two coho had their adipose fins clipped and were observed together upstream of the large pool. Overall conditions during both surveys were favorable as visibility was high and water was clear. Throughout the surveys, several rainbow trout were observed, ranging in length from 80-140mm. The results compare well to a separate survey conducted in 2000 (Benvar 2000) during the coho spawning season which resulted in the observation of 8 spawners believed to be strays from another system. Overall, the results are quite low when compared to nearby indicator streams such as Kirby Creek with Coho returns averaging 225 individuals (range 146-323) over 6.4km of river (35 individuals/km).

Table 4. Spawning Assessment Summary

Task/Date	Target	Location	Staff Gauge (m)/Discharge (cms)	Water Visibility/Stage	Fish Observed	Other Comments
Coho Survey 1 30/09/05	Coho Salmon	Reach 1	1.76m/	Good/High	Several RBT Range (80-135mm)	
Coho Survey 2 20/10/05	Coho Salmon	Reach 1	1.59m/	Good/Moderate	15 RBT Range (80-140mm) – 2 coho observed	Rained previous two days – one coho (wild female) with exasperated caudal fin
Coho Survey 3 03/11/05	Coho Salmon	Reach 1	1.85m/	Good/High	2 coho observed – one in a seal's mouth at slouce	Lots of rain – 2 eagles observed – seal and eagle observations suggest maybe more than two individuals
Coho Survey 4 17/11/05	Coho Salmon	Reach 1	1.58/	Excellent/Low	Several RBT Range (80-135mm)	

Coho Survey 5 15/12/05	Coho Salmon	Reach 1	1.38/	Excellent/Low	Several RBT Range (80-145mm)	
Steelhead Survey 1 02/05/06	Steelhead	Reach 1	1.35/	Excellent/Low	Several RBT Range (80-155mm)	
Steelhead Survey 2 11/05/06	Steelhead	Reach 1	1.34/	Excellent/Low	Several RBT Range (80-145mm)	
Steelhead Survey 3 31/05/06	Steelhead	Reach 1	1.35/	Excellent/Low	Several RBT Range (80-145mm)	

Figure 4. Spawning Assessment Map



3.3 Incubation Assessment

Average survivability to stage sampled at sites 1, 2, 4 and 5 was 94%. Site #3 (mine tailings site) had 100% mortality. Mean survival to stage sampled for all five incubation sites including Site 3 was 76%. Refer to Table 2 below for individual plate summaries as well as Table 3 for incubation site characteristics. Overall, water depth over the incubation cassettes as well as velocities remained within acceptable ranges at all five sites during each site visit sampling stage development. Temperature readings through the incubation period (January 20th – May 31st 2006) ranged from -1.9 ^{o}C to 17.1 ^{o}C averaging 5.6 ^{o}C. The discharge for the same period ranged from 0.107 cms to 5.58 cms (Hydroid Geoscience Ltd. In prep, 2006). As a result of sampling on the exact date of the lowest average daily discharge recorded (May 2nd, 2006) during the survey period, one can deduct that acceptable water levels remained over all incubation sites during the incubation period as all sites were assessed during each incubation site visit.

Water velocity, intragravel oxygen as well as overall incubation conditions for each site appeared favorable at the time of installation. Although site #3 resulted in 100% egg mortality, conditions there remained favorable throughout the assessment period with the exception of the deposition of fines near the outlet of the mine slough tributary (<3m downstream of the incubation site). In comparison, results from a previous incubation assessment conducted in 2002 (Wright and Guimond, 2003) at the exact location appear to be similar. In the 2002 assessment, low survival was attributed to the site having the lowest intragravel dissolved oxygen (mean: 8.25 mg/l, range: 6.68 –10.0) as well as having a heavy deposition of fines and sand over the cassettes. In the 2005 assessment, the intragravel dissolved oxygen was also the lowest of the five sites and silt deposition within the cassettes was also evident.

Results from these two assessments, indicate that low intragravel dissolved oxygen levels caused by silt deposition is limiting growth and stage development at site #3. The deposition is due to two factors:

(1) Backwatering effects: As the site and general area are adjacent to a large pool, silt settles in the relatively slack water away from the main push of the river (right bank).

(2) Local sedimentation: adjacent to site #3, a tributary stream appears to be depositing sediment along the left bank.

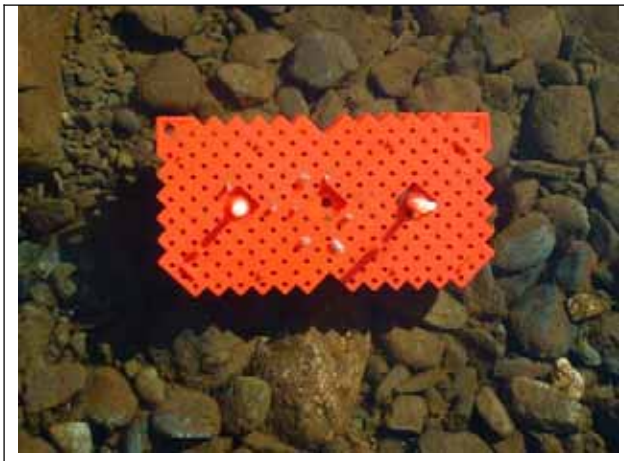
The combination of the two sediment influxes and the lower water velocity eventually reduces the total dissolved oxygen in the gravel. The lower dissolved oxygen is critical to embryonic development (hatch stage) when overall oxygen demand is at its highest. Another factor that might potentially impact development is copper contamination coming in from the left bank tributary. Generally, it was assumed that copper (concentrations considered lethal to fish - 40 times the maximum criteria of 2 µg/l (Wright & Guimond 2003) for freshwater aquatic life (BC Water Quality Guidelines)) bound with various chemicals in the river thereby reducing its lethality. The point at which the copper binds and becomes less lethal is paramount at site #3 as it is possible that the binding/dilution effects may not occur immediately.

3.3.1 Incubation Site Summary

The following incubation site descriptions and summary are detailed below in order of date sampled. Refer to Table 5 (Incubation Assessment Summary 2006) and Table 6 (Incubation Assessment Summary of Pink fry incubation assessment in 2003 – Wright and Guimond 2003) following the site descriptions for detailed site/cassette incubation results.

Site #1 was sampled on March 3rd 2006. Mean survival was 97% (range: 95%-99.5%) up to the hatch stage. Overall, most of the individuals had not yet hatched, however, were still alive. The ATU's at the time of assessment were assumed to be approximately 456 ATU, however, as daily temperature readings for the river were unavailable, the exact number of ATU's at time of assessment was not known. As a result of eggs not being to the stage required, the next site to be assessed was put back for a week to allow for further accumulations of ATU's. Refer to photographs below for typical photograph of stage development.

Site #2 was sampled on May 3rd 2006. Mean survival was 93% (range: 90.5%-96%) up to the emergence stage of development. The ATU's at the time of assessment were assumed to be approximately 821 ATU. Refer to photographs below identifying fry trapped in escape holes as well as a two headed fry.



Sites 3 and 4 were both pulled on April 13th 2006. The ATU's at time of assessment were assumed to be approximately 691ATU's. Mean survival at Site #3 was 0% with all 600 eggs dead. Refer to select photographs below of stage development.

Site #5 was sampled on March 10th 2006. Mean survival was 97% (range: 96%-97%) up to the hatch stage of development. The ATU's at the time of assessment were assumed to be approximately 491 ATU. Refer to photograph below of stage development.

Table 5. Incubation Assessment Summary 2006

Site	*Date Sampled	Plate #1	Plate #2	Plate #3	Mean Survival	Assumed ATU	Actual ATU
1	03/03/06	6 alevins (3%) 1 mort (0.5%) 193 unhatched (96.5%)	6 alevins (3%) 10 morts (5%) 184 unhatched (92%)	9 alevins (4.5%) 6 morts (3%) 185 unhatched (92.5%)	97%	456	348
2	02/05/06	161 emerged (80.5%) 20 unhatched (10%) 3 morts at egg stage (1.5%) ***16 morts (8%)	183 emerged (91.5%) 4 unhatched (2%) 2 morts at egg stage (1%) **11 morts (5.5%)	184 emerged (92%) 8 unhatched (4%) 3 morts at egg stage (1.5%) **5 morts (2.5%)	93% **99%	821	631
3	13/04/06	200 morts (100%)	200 morts (100%)	200 morts (100%)	0%	691	503
4	13/04/06	180 at emergence (90%) 10 emerged (5%) 6 morts at egg stage (3%) **4 morts (2%)	171 at emergence (85.5%) 25 emerged (12.5%) 4 morts at egg stage (2%)	182 at emergence (91%) 15 emerged (7.5%) 2 morts at egg stage (1%) **1 mort (0.5%)	97% **98%	691	503
5	10/03/06	190 alevins (95%) 6 morts (3%) 4 unhatched (2%)	189 alevins (95%) 8 morts (4%) 3 unhatched (1%)	188 alevins (94%) 6 morts (3%) 6 unhatched (3%)	97%	491	367

* - eggs put into gravel at 256 ATU on January 19th, 2006

** - Mean Survival Includes morts trying to emerge
*** - Dead trying to emerge – caught in escape hole.

Table 6. Incubation Assessment Summary (Pink Salmon) – Wright and Guimond 2003

Site # (2003)	Site # in 2006 assessment	Plate #1 Mean Survival %	Plate #2 Mean Survival %	Plate #3 Mean Survival %	Plate #4 Mean Survival %	Plate #5 Mean Survival %	Plate #6 Mean Survival %	Overall Site Survival % 2003	Overall Site Survival % 2006
1	2	3.5	12	2	11.5	23.5	35.5	14.7	93
2	3	0	0	20.5	0	15	10.5	7.7	0
3	4	94.5	43	99	99	98	Lost Cassette	86.7	97

3.4 Spawning Gravel Assessment

Spawning gravel in reaches 1 and 2 was generally considered to be of low-moderate quality ranging in size from 20mm to 105mm interspersed between larger substrate. Total gravel area for Reaches 1 and 2 totaled 144m^2 of which 119m^2 (82.6%) was located in Reach 1 and 25m^2 (17.4%) located in Reach 2 (refer to Figure 5 and 6 below for spawning gravel locations). Present spawning gravel locations are primarily within the wetted channel at low/moderate flow conditions, that being discharges ranging from 0cms to 13.1cms according to Figure 2 above. The total area available within these low/moderate flow conditions is 118m^2 (82%) out of the total area of 144m^2. As a result, an additional 18% of the gravel is available outside of low/moderate flows. Of the 18% outside of the low/moderate water discharge wetted area, another 18m^2 (12.5%) is available to spawners when flows are higher, typically within November to March (16.4cms to 31.1cms). The resulting 8m^2 (5.5%) would be available to flows considered high to extremely high (>227 MAD). The planned flow release of 0.25cms could therefore potentially expand on the additional 5.5% resulting in a net increase of 8m^2.

Figure 5. Spawning Gravel Assessment Map

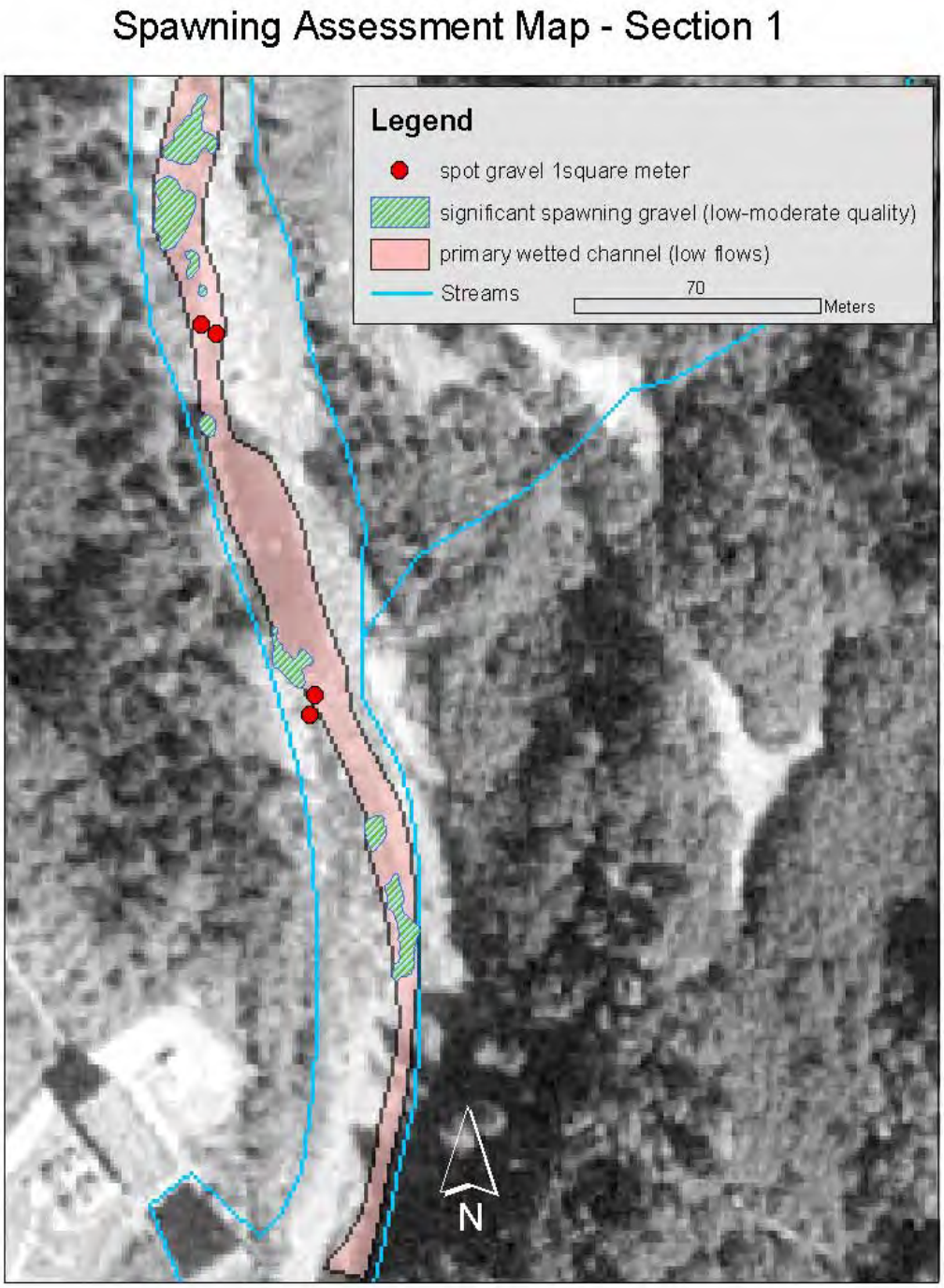
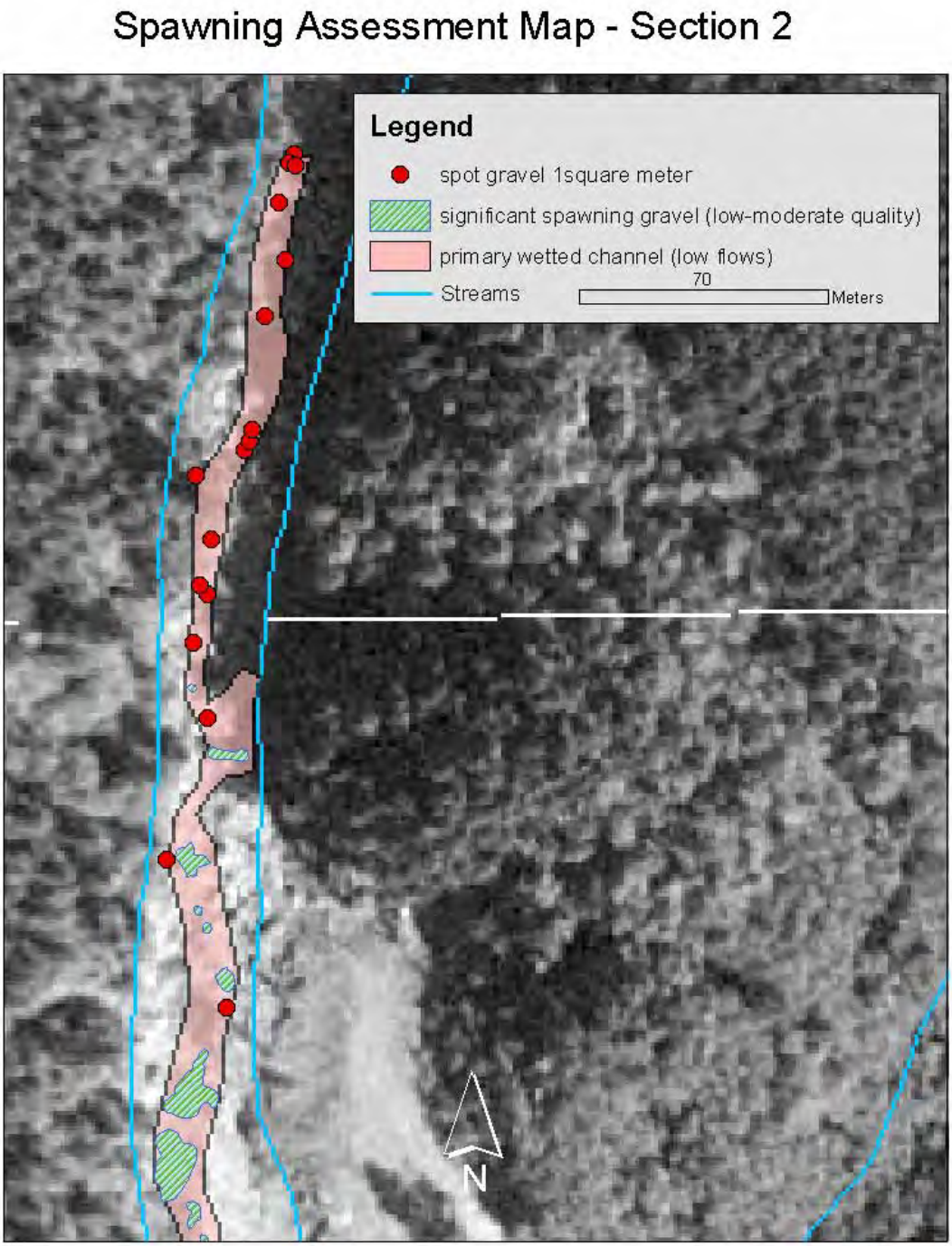


Figure 6. Spawning Gravel Assessment Map



4.0 Summary and Discussion

Overall, only 4 adults (coho) were observed during our assessments of which two were hatchery fish. It is speculated that low returns may be due to several limiting factors including high adult mortalities associated with predation (i.e seal, eagles etc.), water quality (point source pollution from a decommissioned copper mine along the left bank of Reach 1), as well as minimum spawning habitat availability/quality. Of the limiting factors, water quality concerns (high copper concentrations) appear to be limited to a small section of river, where an unnamed tributary comes in through mine tailings before being deposited into the left bank. Overall, spawning gravel area was limited to 144 square meters of which 118 square meters is withinlow/moderate discharge levels. As a result of a potential spawning area of 144 m^2, the carrying capacity for coho redds is 51 based on an average redd area of 2.8 m2 (Reiser and Bjornn 1979), and *Habitat Suitability Index Models and Instream Flow Suitability Curves: Coho Salmon* (McMahon 1983). Alternatively, the carrying capacity for the number of adults this system would support is 12 pairs based on 1 spawning pair for every 12 m^2 (Groot and Margolis 1991).

The egg incubation assessment revealed that overall mean survival was high (94%) at 4 out of the 5 sites with the exception of Site #3 adjacent to the mine tailings, which had 100% mortality. It is hypothesized that the planned flow release of 0.25 cms will not substantially increase spawning habitat, but may improve on dissolved oxygen within existing spawning habitat and therefore increase egg/fry survivability at select sites within the river.

The data collected for this project represents year 1 of a two year pre-treatment phase before 0.25cms flow release will be initiated in Jun 2007. After two years of pre-treatment data, four more years of monitoring will be conducted. After all six years of data are collected, an in-depth analysis of factors limiting spawning success in the Lower Jordan River will be conducted. It is premature to define factors limiting production at this initial stage in the monitoring program.

5.0 Recommendations

This study encountered several issues associated with the implementation of the study terms of reference that should be rectified in year two to increase the effectiveness. The following improvements to the study program are recommended:

- Egg incubation assessment protocol: a protocol for handling, transferring and assessing eggs during the incubation assessment process should be developed and reviewed by BC Hydro and hatchery managers prior to implementation in early 2007. This will ensure that replicates will be installed correctly, hatchery communications and requirements are understood ahead of time, and handling issues are mitigated.
- Flow records: flow records for the study period were not available at the time of report development and represent a key component of spawning success assessment. Access to flow records for the period of study is required to document any flow related impacts to spawning observations and incubation assessments made as part of this study.
- Broaden steelhead assessment period: all steelhead surveys were completed in May 2006, and it is recommended that surveys in the future include both earlier and later observations in consideration of steelhead run timing in adjacent watersheds under observation.

6.0 References

BC Hydro. 2001. JOR WUP data collection terms of reference.

BC Hydro. 2005. Jordan River Water Use Plan Monitoring Program Terms of Reference.

Griffith, R.P. 1996. Biophysical Assessment of fish production within the Jordan River drainage 1994. Final Report. R.P. Griffith and Associates, Sidney, BC for BC Hydro and Power Authority, Burnaby, BC.

Hydroid Geoscience Ltd. *in prep*. Lower Jordan River Inflow Monitoring Study. Prepared for BC Hydro, Burnaby, BC.

Reiser, D.W., and T.C. Bjornn. 1979. 1. Habitat Requirements of Anadromous Salmonids. W.R., Technical Editor. Influence of Forest and Rangeland Management on Anadromous Fish Habitat in the Western United States and Canada. USDA Forest Service GTR PNW-96. 54 pp.

Wright, M. C. and E. Guimond, 2003. Jordan River pink salmon incubation study. Prepared for the Bridge-Coastal Restoration Program, Burnaby, BC.